

## Theory of Computation

CSIT — 4th Semester — 2082 — Answer Sheet  
Subject Code: CSC262

**Note:** This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

### Section A

**1. List the application of Pumping Lemma. Minimize the following DFA using Table Filling Algorithm.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**2. Define yield in parse tree with example. Convert the following grammar to CNF** $(A \rightarrow AS \mid A \mid B \rightarrow S \mid S \mid A \mid 1 \mid 1)(S \rightarrow \epsilon)$

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**3. Distinguish between multi - tape and multi - track Turing Machine. Design a TM that accepts a string over alphabet  $\Sigma = \{a, b\}$ , such that  $L = \{a^n b^n \mid n > 0\}$ .**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

### Section B

Attempt any Eight questions[8×5=40]

**4. Why do we need asymptotic notation? Discuss about alphabet and power of alphabet.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**5. Define ambiguous grammar. How do you use parse tree to show the ambiguity of a grammar?**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**6. Given the non-deterministic Turing machine,  $(TM: (\{q_0, q_1, q_2, q_3\}, \{0, 1\}, \{0, 1, B\}, \delta, q_0, B, \{q_3\}))$  with the following transition rules, describe the language accepted by given Turing machine.** $(\delta(q_0, 0) = \{(q_0, 1, R), (q_1, 1, R)\}; \delta(q_1, 1) = \{q_2, 0, L\}; \delta(q_2, 1) = \{q_0, 1, R\}; \delta(q_1, B) = \{q_3, B, R\})$

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

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**7. Given two DFAs  $(M_1)$  and  $(M_2)$  over the same alphabet  $(\Sigma)$ , design a third DFA  $(M_3)$  such that  $(M_3)$  will accept the string  $(w \in \Sigma^*)$  if it is accepted by both  $(M_1)$  and  $(M_2)$ .**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**8. Differentiate between Moore and Mealy machines.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**9. Design a DFA that accepts the odd length of binary strings**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**10. How do you convert CFG to PDA? Explain.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**11. Why do we need  $(\epsilon)$  transition? Give the formal definition of PDA.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**12. Describe about abstract, decision and optimization problems.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

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*These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.*